Data Architectures for Health under the LGPD: Institutional Coordination, Risk, and Governance in Brazil

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Abstract

The rapid expansion of health data infrastructures in Brazil has intensified demands for interoperability, benchmarking, and evidence-based policymaking. At the same time, the Brazilian General Data Protection Law (LGPD), reinforced by the constitutional recognition of data protection as a fundamental right under Constitutional Amendment 115/2022, has redefined the institutional conditions under which sensitive health data may be processed and shared. This article examines which data architectures are legally and institutionally sustainable in Brazil, given the structural fragmentation of public health databases, the coexistence of multiple data entry points—individuals, health services, disease surveillance programs, and private digital platforms—and the growing interaction between public and contract-based data ecosystems.

The central argument is that centralized models of large-scale health microdata integration are normatively fragile and institutionally costly under the Brazilian constitutional and data protection framework. In response, the article proposes a federated,

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risk-based, layered architecture that embeds purpose limitation, necessity, proportionality, and accountability directly into system design. The proposed model combines: (i) retention of identifiable data at their institutional source; (ii) controlled access to de-identified microdata within secure processing environments; (iii) distributed and privacy-preserving analytical coordination; and (iv) public dissemination of aggregated and reproducible outputs.

Methodologically, the paper adopts a design-oriented institutional approach. In addition to legal-constitutional analysis and systematic mapping of Brazil's health data ecosystem, it examines recent regulatory documents issued by the Brazilian Data Protection Authority (ANPD) between 2023 and 2024 as empirical evidence of institutional incorporation of constitutional data protection principles. These documents allow the article to move beyond doctrinal abstraction and identify how risk-based reasoning is being operationalized in practice. Comparative governance literature—including the United Kingdom's Trusted Research Environment model—grounds the proposal in international experience while preserving the specificity of the Brazilian constitutional and federative context.

1 Introduction

The production and circulation of health data have become central to public policymaking, empirical research, and institutional evaluation in Brazil. Over recent decades, the Brazilian state has consolidated a complex ecosystem of public health databases, combining national surveys, administrative records of the Unified Health System (SUS), and epidemiological surveillance systems. Simultaneously, digital platforms and mobile applications have generated new streams of privately produced health-related data structured through contractual adhesion, creating a heterogeneous informational landscape that operates under distinct legal and institutional logics.

This expansion has intensified pressure for interoperability and integrated evidence production. Yet the enactment of the LGPD—reinforced by the elevation of data protection to the status of a fundamental right through Constitutional Amendment 115/2022—has reshaped the institutional conditions of data use. Health data systems can no longer be conceived as neutral repositories to be optimized primarily for analytical access. They must operate as governance architectures that embed privacy, traceability, security, and institutional coordination from inception.

This article argues that centralized integration of health microdata is neither legally sustainable nor institutionally efficient under the Brazilian constitutional framework. The argument proceeds in three stages. Section 2 maps the structural features of Brazil's health data ecosystem, identifying heterogeneous units of analysis, access regimes, and risk profiles. Section 3 examines the LGPD as an architectural constraint, highlighting how constitutional proportionality and regulatory guidance shift interoperability away from maximal integration and toward governed coordination. Section 4 proposes a federated, risk-based architecture, clarifying the roles of access nodes, privacy-preserving analytical coordination, and accountability mechanisms. Section 5 concludes by situating the proposal in broader debates on data-intensive governance and constitutional democracy.

Methodologically, the paper adopts a design-oriented institutional approach. It combines legal-constitutional analysis, systematic mapping of Brazil's health data ecosystem, and an applied policy evaluation scenario to identify feasible arrangements under existing legal

constraints. The analysis draws on comparative governance literature—including the United Kingdom's Trusted Research Environment model—to ground the proposal in international experience without displacing the specificity of the Brazilian constitutional and federative context.

2 Brazil's Health Data Ecosystem: Layers, Tensions, and Governance Challenges

Brazil's health data ecosystem is marked by structural heterogeneity that is neither accidental nor purely technical. It reflects decades of institutional layering within a federative public health system in which federal, state, and municipal actors exercise overlapping yet distinct responsibilities. As a result, the SUS generates multiple information systems designed for different administrative, financial, and epidemiological purposes. Understanding this ecosystem therefore requires more than a descriptive inventory of databases; it demands attention to how unit of analysis, update cycles, custody arrangements, and legal mandates interact within a multi-level governance structure.

Table 1 highlights these differences. National surveys such as the *Pesquisa Nacional de Saúde* (PNS) produce probabilistic, population-level evidence and are periodically released as anonymized microdata. By contrast, administrative systems within DATASUS—including SIH, SIA, and CNES—generate continuous operational records that may contain longitudinal and potentially identifiable traces. Nutritional surveillance systems such as SISVAN retain sensitive clinical data under strict institutional custody, while epidemiological surveillance initiatives combine aggregated dissemination with controlled microdata access. Private digital platforms, operating under contractual adhesion rather than statutory mandate, generate persistent, high-frequency individualized records.

These differences have immediate legal significance. Under a proportionality-based framework, heterogeneity shapes the boundaries of permissible linkage. Although survey data are anonymized, their disclosure controls may become fragile when combined with granular administrative sources. Reidentification risk increases non-linearly as cross-dataset

linkage becomes technically feasible (*Health Data Governance: Privacy, Monitoring and Research* 2017). In the Brazilian context—where administrative microdata are increasingly available—linkage can restore identifiability in ways that neither dataset would permit independently.

Administrative systems intensify this concern. Originally designed for reimbursement and monitoring rather than integrated analytics, they exhibit fragmentation that complicates interoperability but simultaneously limits systemic exposure. Longitudinal traces and geographic granularity—especially in small municipalities—amplify disclosure risk. When multiple administrative layers are combined, exposure becomes systemic rather than localized. Consequently, mitigation cannot rely solely on procedural safeguards; it must be embedded architecturally.

Surveillance systems introduce further complexity. While aggregated outputs are publicly disseminated, underlying microdata capture behavioral and epidemiological patterns that may alter their risk profile when recombined with other sources (Kaye and al. 2018). At the same time, private digital platforms operate under contractual consent regimes that differ fundamentally from statutory public-interest mandates. Consent obtained in commercial contexts does not automatically translate into a valid legal basis for public-sector analytics, particularly when health data are classified as sensitive under the LGPD.

The software infrastructure within the federal public health system is predominantly characterized by profound technological obsolescence and the absence of a cohesive enterprise data architecture. When examining the operational reality of State Health Secretariats (SES), it becomes evident that these agencies rely heavily on monolithic legacy systems. These platforms were developed without the application of modern architectural standards, rendering interoperability across municipal levels a structural challenge of immeasurable proportions in the current landscape.

This architectural deficit results in applications plagued by low scalability and high latency. Consequently, the outdated interfaces of these systems impose a severe cognitive load on healthcare professionals, demanding exhaustive and redundant data entry routines across multiple disconnected environments. This scenario generates not only the duplication of operational procedures but, more importantly, the discontinuity of clinical care delivery.

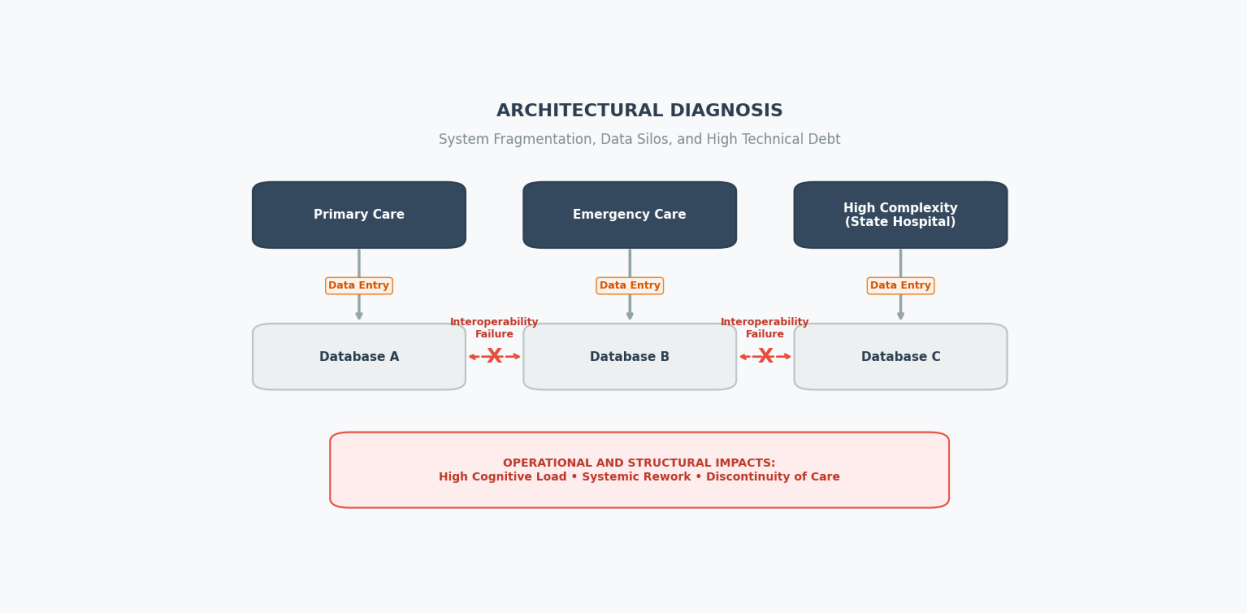

Figure 1: Architectural Diagnosis

Regarding information sharing, the landscape reflects a critical deficit in both semantic and syntactic interoperability. The lack of native standardization in health integration protocols acts as the primary catalyst for the proliferation of isolated institutional data silos .

To provide a pragmatic illustration of the severity of this structural scenario, one need only observe the coordinating role of the State Secretariats themselves. In practice, the healthcare network operates on fundamentally incompatible database logics: a regional hospital might record a patient's admission in a proprietary system "A," while the municipal Emergency Care Unit (UPA) that initiated the transfer utilizes system "B," with both operating under entirely distinct primary keys and data dictionaries.

Without a native integration layer or a standardized data bus, the SES is entirely incapable of cross-referencing this information in an automated manner. This structural bottleneck necessitates a massive human and computational effort to extract, cleanse, and unify the records, ultimately rendering real-time interoperability technically and financially unfeasible under the current framework.

These dynamics converge in what may be described as a *fragmentation paradox*. Fragmentation is often framed as a coordination deficit; yet, from a risk-based perspective, it functions as a structural buffer against systemic exposure. The governance challenge, therefore, is not

Table 1: Structural Characteristics of Major Brazilian Health Databases

Database type	**Primary unit of analysis**	**Sensitivity / identifiability**	**Update cycle**	**Typical access regime**
National health surveys (e.g., PNS)	Individual (sampled)	Medium (anonymized microdata; residual linkage risk)	Periodic (multi-year)	Public microdata with statistical controls
Administrative systems (SIH, SIA, CNES)	Procedure / attendance / establishment	High (potentially identifiable records; longitudinal traces)	Continuous	Restricted; formal authorization; institutional custody
Nutritional surveillance (SISVAN)	Individual / clinical record	High (sensitive longitudinal health data)	Continuous	Institutional access only; strong purpose limitations
Epidemiological surveillance (e.g., Vigitel)	Individual (survey-based)	Medium (microdata controlled; aggregated dissemination)	Annual	Aggregated public release; controlled microdata
Territorial indicators (coverage, financing)	Municipality / region	Low	Annual	Open data
Private digital platforms	Individual (persistent ID)	High (behavioral + health; secondary uses common)	Real-time	Contract-based, private control

to eliminate fragmentation by default but to transform it into a managed form of coordination consistent with constitutional constraints. This insight becomes even more salient when viewed through Brazil's federative structure, summarized in Table 2. Custody, authority, and interoperability capacity are constitutionally distributed. Any sustainable data architecture must therefore operate within this distributed arrangement rather than attempt to override it through centralization.

The governance challenges observed in Brazil mirror patterns identified in comparative health data scholarship. The OECD (*Health Data Governance: Privacy, Monitoring and Research* 2017) identifies fragmentation, interoperability deficits, and privacy risk escalation as recurrent challenges across national health systems, increasingly favoring federated governance models that preserve institutional autonomy while enabling controlled analytical coordination. In the United Kingdom, Trusted Research Environments (TREs) operationalize this logic by

Table 2: Data Governance and Access Structures Across Federative Levels

Level	**Typical role in data governance**	**Control over identifiable data**	**Interoperability capacity**
Federal (Ministry of Health)	Normative coordination; system standardization; national databases	Partial (central systems; shared governance with subnational levels)	High (standard-setting authority)
State secretariats	Regional coordination; hospital network management	Yes (regional records)	Medium–High
Municipal secretariats	Primary care and local service management	Yes (local clinical records)	Medium
Public hospitals	Clinical record custodianship	Yes (medical records; professional secrecy)	Low–Medium
Private providers	Service delivery; contract-based data retention	Yes (private records)	Variable

keeping data in secure custody while granting controlled analytical access (*Trusted Research Environments (TREs): Principles and Best Practices* 2021). The biomedical research literature underscores the limits of consent-based frameworks and calls for institutional accountability mechanisms to govern sensitive data when consent is impracticable (Kaye and al. 2018; Rumbold and Pierscionek 2017).

Brazil's federative SUS structure adds complexity that makes direct transplantation impossible. Distributed authority may enhance democratic accountability but complicates interoperability (Arretche 2012; Lima and Viana 2011). The Brazilian case is therefore a distinctive laboratory for examining how constitutional data protection interacts with multi-level public administration, making both purely technical and purely legalistic approaches insufficient.

3 The LGPD as Architectural Constraint: Constitutional Law and Structural Reconfiguration

The LGPD establishes a general regime for personal data processing applicable to both public and private actors. In the health sector, however, its implications extend beyond compliance obligations and into the domain of institutional design. Because health data are classified as

sensitive, heightened safeguards apply across the processing lifecycle. As the Brazilian Data Protection Authority (ANPD) emphasized in its 2023 Technical Study on Anonymization (Autoridade Nacional de Proteção de Dados 2023a), anonymization must be understood as contextual and dynamic rather than as a one-time technical procedure. Reidentification risk depends on available auxiliary information, technical capabilities, and adversarial context, a view consistent with broader data governance scholarship (*Health Data Governance: Privacy, Monitoring and Research* 2017). In an evolving digital ecosystem, architectures that treat anonymization as permanent rather than contingent are inherently unstable.

This architectural reading gains further support from constitutional jurisprudence. In ADI 6387 (2020), the Supreme Federal Court made clear that broad circulation of personal data—even in contexts of compelling public interest—must satisfy necessity and proportionality requirements. The decision functions less as a case-specific ruling and more as an institutional signal: governance strategies premised on unrestricted sharing or large-scale integration heighten constitutional vulnerability regardless of policy objectives. The subsequent constitutionalization of data protection through Constitutional Amendment 115/2022 consolidates this shift, embedding informational self-determination within the structure of fundamental rights and transforming proportionality into a design constraint.

From a theoretical perspective, data protection thus operates as constitutional design logic (Floridi 2018; Hildebrandt 2020). Information infrastructures embody normative commitments: choices regarding storage, access, interoperability, and accountability reflect constitutional values (Mendes 2021; Bioni 2020). Under this framework, interoperability cannot be equated with maximal integration. Instead, it must be reconceptualized as legally structured coordination across distributed institutional nodes.

Importantly, this transformation is not confined to doctrine. Between 2023 and 2024, the ANPD issued technical studies that operationalize risk-based reasoning in concrete governance contexts. The Technical Study on Anonymization conceptualizes de-identification as a structured, iterative process (Autoridade Nacional de Proteção de Dados 2023a, pp. 4–7). The Regulatory Sandbox framework introduces supervised experimentation grounded in proportionality and mitigation mechanisms (Autoridade Nacional de Proteção de Dados 2023b, pp. 4–7). The Technological Radar on Biometrics identifies systemic risks affecting vulnerable

populations and underscores the continued applicability of general data protection principles (Autoridade Nacional de Proteção de Dados 2024, pp. 6, 9). As synthesized in Table 3, these documents reveal a regulatory trajectory toward risk-sensitive governance. Yet they remain sectoral rather than architecturally integrated, leaving a gap between constitutional principle and infrastructural consolidation.

3.1 Institutional Responses to Constitutional Data Protection (2020–2024)

The constitutional recognition of data protection as a fundamental right, reinforced by Constitutional Amendment No. 115/2022 and interpreted by the Supreme Federal Court in ADI 6387 (2020), signaled that state data practices must be structured through proportionality, necessity, and risk-based reasoning. To assess whether this constitutional signal has translated into institutional practice, this section examines technical documents issued by the Brazilian Data Protection Authority (ANPD) between 2023 and 2024.

Three documents are particularly relevant: (i) the Technical Study on Anonymization (2023), (ii) the Regulatory Sandbox Technical Study (2023), and (iii) the Technological Radar on Biometrics and Facial Recognition (2024).

Risk-Based Anonymization and the Limits of Definitive De-Identification The ANPD's Technical Study on Anonymization explicitly frames anonymization not as a static condition, but as a process dependent on contextual risk. The document emphasizes that anonymization must be assessed considering "meios técnicos razoáveis e disponíveis na ocasião do tratamento" and acknowledges that re-identification risks are dynamic and dependent on auxiliary information(Autoridade Nacional de Proteção de Dados 2023a, pp. 4–7).

Importantly, the study conceptualizes anonymization as a structured process involving successive technical steps rather than a binary state of compliance. It also highlights the phenomenon of "combinações únicas" and quasi-identifiers, noting that even indirect attributes may enable re-identification when combined(Autoridade Nacional de Proteção de Dados 2023a, pp. 6–7).

This institutional interpretation reinforces a central architectural implication: if anonymization is inherently contextual and reversible under certain technical conditions, governance cannot rely on centralized repositories presuming permanent de-identification.

Experimental Regulation and Institutional Learning The ANPD's Technical Study on Regulatory Sandboxes adopts a "test and learn" regulatory philosophy aligned with OECD guidance(Autoridade Nacional de Proteção de Dados 2023b, pp. 4–6). The document defines sandboxes as structured environments that allow innovative technologies to be tested under regulatory supervision, emphasizing proportionality and risk mitigation mechanisms.

The study stresses that sandboxes are not deregulatory tools but instruments of "descoberta regulatória," aiming to understand how innovations interact with existing legal frameworks(Autoridade Nacional de Proteção de Dados 2023b, pp. 6–7). This emphasis on structured experimentation suggests that institutional design, rather than reactive enforcement, is central to effective governance in high-risk technological domains.

Such reasoning is directly applicable to health data governance: rather than permitting broad data circulation followed by ex post correction, regulatory architecture should embed safeguards and supervision ex ante.

Biometric Data and Structural Risk Recognition The 2024 Technological Radar on Biometrics and Facial Recognition further illustrates institutional awareness of systemic risk. The document underscores that biometric data are sensitive and may affect "massas populacionais, inclusive grupos vulneráveis"(Autoridade Nacional de Proteção de Dados 2024, p. 6). It also recognizes risks of discriminatory outcomes, false positives, and lack of transparency in data sharing practices.

Notably, the study acknowledges the absence of comprehensive federal regulation specifically addressing facial recognition while stressing that even in contexts excluded under Article 4 of the LGPD (such as public security), general data protection principles remain applicable(Autoridade Nacional de Proteção de Dados 2024, p. 9).

The document concludes by recommending further study in areas such as secondary data sharing and biometric data processing in educational settings, signaling that institutional

risk perception extends beyond narrow compliance concerns.

Fragmented Incorporation and Architectural Implications Taken together, these documents reveal that constitutional data protection principles are being progressively internalized by the Brazilian regulatory authority through risk-based reasoning, proportionality analysis, and experimental governance instruments.

However, the institutional responses remain programmatic and sectoral rather than structurally integrated. The absence of a unified architectural framework for public health data means that proportionality and risk assessment are often addressed at the level of individual systems or technologies.

This empirical pattern supports the central claim of this article: constitutional data protection in Brazil is evolving toward a risk-based and design-oriented paradigm, but its effectiveness depends on embedding these principles within the architecture of data governance itself. Without a federated and layered model that distributes analytical capacity while minimizing systemic exposure, constitutional guarantees remain unevenly operationalized. Table 4 synthesizes how recent ANPD documents operationalize constitutional data protection through risk-based reasoning and sector-specific guidance. While these instruments demonstrate progressive institutional internalization of proportionality and accountability, they remain fragmented across thematic domains. This fragmentation reinforces the need for a coherent architectural framework capable of translating constitutional principles into systemic data governance design.

4 A Federated, Risk-Based Architecture: Design, Governance, and Institutional Roles

If constitutional proportionality now functions as an architectural constraint, the question becomes how to translate this constraint into institutional form. The convergence of constitutional doctrine, emerging regulatory practice, and comparative governance experience points toward a federated, risk-based, layered architecture as the most viable response to Brazil's health data governance challenge. Rather than treating interoperability as a technical

Table 3: Institutional Incorporation of Constitutional Data Protection Principles (2023–2024)

Document (ANPD)	**Core Governance Focus**	**Risk-Based Elements Identified**	**Architectural Implication for Health Data**
Technical Study on Anonymization (2023)	Definition of anonymization as contextual and process-based	Dynamic re-identification risk; quasi-identifiers; "reasonable means" standard	Centralized repositories cannot presume definitive de-identification; governance must minimize systemic exposure
Regulatory Sandbox – Technical Study (2023)	Experimental and supervised regulatory environments	Proportionality; structured mitigation mechanisms; regulatory learning	Data governance should embed ex ante safeguards rather than rely on ex post enforcement
Technological Radar – Biometrics and Facial Recognition (2024)	Assessment of biometric technologies and systemic impacts	Sensitive data classification; discriminatory bias; transparency gaps; mass population effects	High-risk technologies require layered supervision and controlled data circulation

objective—moving data across systems—the model reconceives it as an institutional strategy: coordinating analysis under rights constraints.

As operationalized in Table 4, the architecture differentiates nodes according to custody, access conditions, and risk level. Identifiable records remain at their institutional source under clinical or administrative control. Secure analytical environments permit purpose-bound access to de-identified microdata without enabling raw export. Federated query nodes allow cross-institutional modeling through execution near the data rather than through data transfer. Public dissemination nodes release aggregated outputs, while oversight nodes ensure supervisory access to logs and governance metadata.

This structure embeds proportionality directly into infrastructure: as data move across nodes, exposure decreases. Privacy-preserving computational paradigms such as federated learning reinforce this alignment by enabling inference without centralization (Yang and al. 2019; Rieke and al. 2020). Their importance lies not merely in technical innovation but in

Table 4: Access Nodes in the Proposed Federated Architecture

Node type	**Data custody**	**Access conditions**	**Risk level**
Clinical custody node	Identifiable health records	Restricted to primary purpose; internal access only; professional secrecy	High
Secure analytical environment node	De-identified microdata	Controlled, purpose-bound, auditable access; no raw export	Medium
Federated query node	Distributed analytical coordination	No raw data transfer; model execution near data; governed outputs	Low–Medium
Public dissemination node	Aggregated outputs	Open access; reproducibility and transparency	Low
Regulatory oversight node	Audit logs and governance metadata	Supervisory access (ANPD/internal audit); accountability and incident review	Low

normative compatibility: they instantiate data minimization at the computational layer.

An applied illustration clarifies the distinction. A federal evaluation linking SISVAN nutritional outcomes with PNS socioeconomic indicators would, under a centralized model, generate a combined dataset with elevated reidentification risk—particularly in small municipalities. Under the federated model, analytical specifications are coordinated centrally but executed within secure environments at each node, and only aggregated outputs are returned. The architecture therefore redistributes computation rather than restricting evidence production, preserving analytical capacity while reducing systemic vulnerability.

Finally, institutional sustainability depends on governance discipline. Clear delineation of controllers and operators across SUS levels, formalized data-sharing agreements, and mandatory Privacy Impact Assessments transform proportionality from abstract doctrine into auditable design. Security infrastructure—segmented authentication, layered access controls, and logging—provides technical reinforcement. Moreover, distinguishing public statutory mandates from private consent-based ecosystems prevents constitutional standards from being diluted through contractual asymmetry. Interaction with private data sources remains possible, but incorporation must rely on independent legal bases and pass through the same governance layers.

5 Conclusion

Digital infrastructures intensify a fundamental tension between data-driven governance and constitutional safeguards. This article has argued that centralized integration of health microdata is neither normatively required nor institutionally optimal under Brazil's LGPD and constitutional framework. Fragmentation, rather than a dysfunction to be eliminated by default, is a governance condition that can reduce systemic exposure when properly coordinated.

The federated, risk-based architecture proposed here reconciles analytical capacity with constitutional proportionality. By retaining identifiable data at institutional sources, enabling governed access within secure environments, distributing computation through privacy-preserving protocols, and disseminating only aggregated reproducible outputs, the model supports evidence-based health governance at lower legal and institutional risk. More broadly, data protection law should be read not as a constraint on empirical governance, but as a constitutional guide for the design of data-intensive public institutions. In democratic states, the legitimacy of data-intensive governance depends not on maximal integration, but on structured coordination, traceable accountability, and the institutional embedding of fundamental rights.

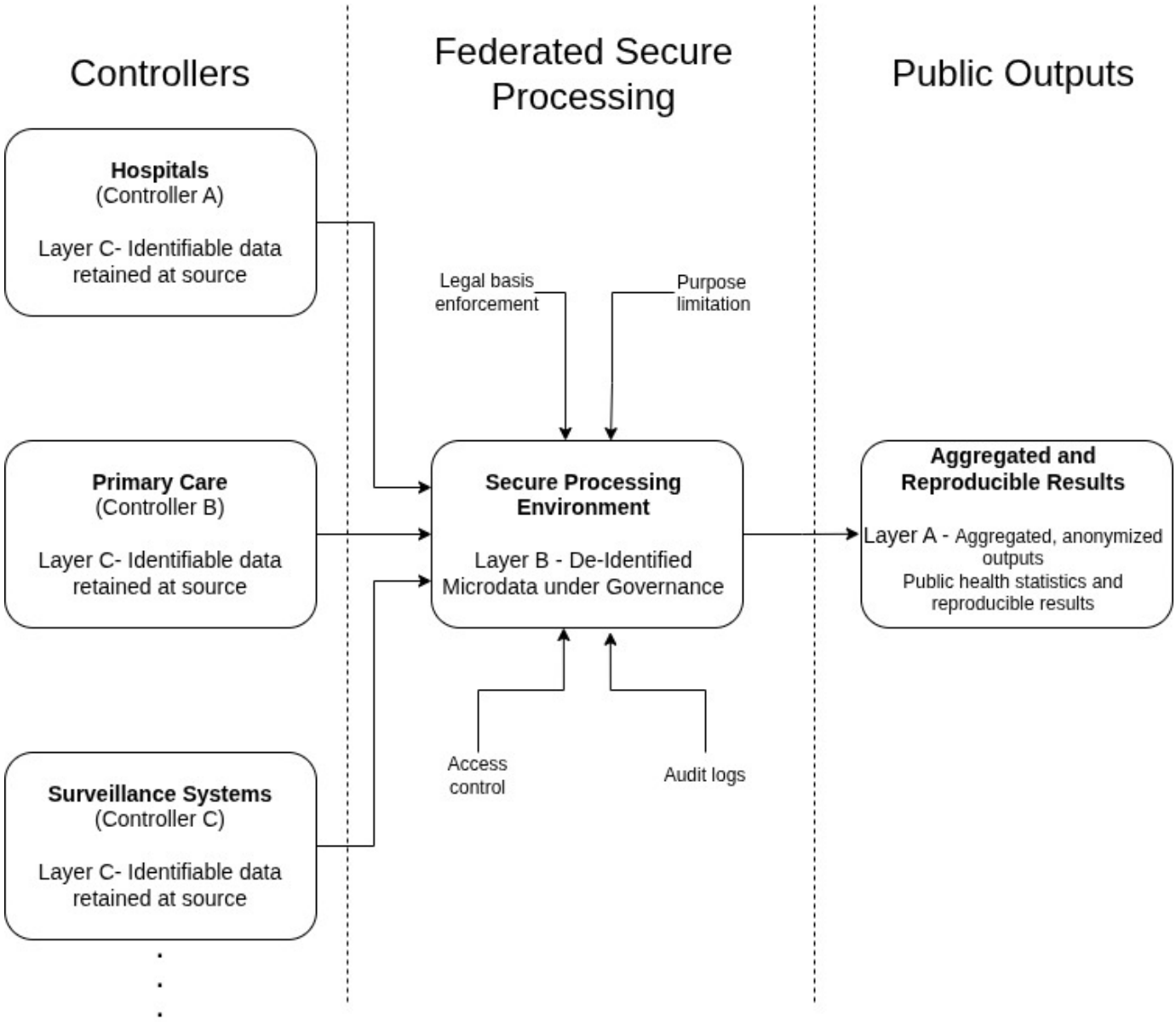

Figure 2: Architectural Model Diagram

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